EDMS 02-201-2

Technical Specification for
24 kV Automated Smart RMU (GIS / AIS)

<u>Indoor/Outdoor Installation</u>

Issue: February - 2026 / Rev- 2

- **<u>This revision contains option items that must be selected by EDC before bidding.</u>**
- **<u>EDMS 19-400 for HRC Fuses shall be attached if applicable.</u>**
- **<u>EDMS 17-400 for MV CTs shall be attached if applicable.</u>**
- **<u>EDMS 17-401 for MV VTs shall be attached if applicable.</u>**
- **<u>EDMS 20-305 for Indirect Smart Meters shall be attached if applicable.</u>**

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GLOSSARY:

EEHC: Egyptian Electricity Holding Company.

EDMS: Egyptian Distribution Material Specifications.

MV: Medium Voltage.

SRMU: Smart Ring Main Unit.

GIS: Gas Insulated System.

AIS: Air Insulated System.

A.C.: Alternating Current.

D.C.: Direct Current.

IEC: International Electrotechnical Commission.

NO: Normal Open.

NC: Normal Close.

VDIS: Voltage Detecting and indicating Systems.

EMS: Energy Management System.

DMS: Distribution Management System.

DA: Distribution Automation.

SA: Substation Automation.

RTU: Remote Terminal Unit.

ADMS: Advanced Distribution Management System.

SCADA: Supervisory Control and Data Acquisition.

C.T.: Current Transformer.

V.T.: Voltage Transformer.

SF6: Sulfur Hexafluoride

IP: Ingress Protection.

IK: Degree of Protection against Mechanical Impact.

C.B.: Circuit Breaker.

L.B.S.: Load Break Switch.

H.R.C.: High Rupture Capacity.

…EDC: …. Electricity Distribution Company.

LSC: Loss of Service Continuity.

IAC: Internal Arc Classification.

AFLR: Access Front Lateral Rear.

LPCT: Low Power Current Transformer

FPI: Fault Passage Indicator.

LED: Light-Emitting Diode.

TCP: Transmission Control Protocol.

O.C.: Over Current.

E.F.: Earth Fault.

S.C.: Short Circuit.

IDMT: Inverse Definite Minimum Time.

DT: Definite Time.

WAN: Wide Area Network.

SCBO: Select Check Before Operate.

SNMPv3: Simple Network Management Protocol.

XLPE: Cross-linked polyethylene.

HMI: Human Machine Interface.

P.C.: Personal Computer.

RTC: Real time Clock.

PS: Power Supply.

GPS: Global Positioning System.

IRIG-B: Inter-Range Instrumentation Group Timecodes.

SNTP: Simple Network Time Protocol.

RBAC: Role-Based Access Control.

SYSLOG: System Logs.

DNP: Distributed Network Protocol.

1. SCOPE

This specification specifies the minimum technical requirements for design, supply, engineering, manufacturing, testing, inspection, packing, and performance of indoor/outdoor gas-insulated (AIS/GIS) smart ring main unit (SRMU) intended to be used in a 24 kV medium-voltage system. This SRMU should be controlled remotely by using integrated communication devices.

2. REFERENCE

The smart RMU equipment should comply with IEC listed hereafter. In no way does this specification supersede the relevant codes except when the requirements herein are more stringent. All cases of conflicts should be immediately brought to the tenderer's attention.

3. APPLICABLE STANDARDS

Unless otherwise specified, the SRMU should comply with the latest edition of IEC publications:

IEC 62271-1	Common Specification for alternating current switchgear and control gear
IEC 62271-200	AC metal-enclosed switchgear and control gear for rated voltage above 1 kV and up to and including 52 kV.
IEC 62271-100	High-voltage alternating-current circuit breakers
IEC 62271-102	Alternating current disconnectors (isolators) and earthing switch.
IEC 62271-103	Switches for rated voltages above 1 kV up to and including 52 kV
IEC 62271-105	Alternating current switch-fuse combinations for rated voltages above 1 kV up to and including 52 kV.
IEC 62271-213	High-voltage switchgear and control gear - Part 213: Voltage detecting and indicating system
IEC 60870 – 5/6	Tele control equipment and system (EMS; DMS; DA; SA).
IEC 61131-5	Programmable controllers – communication.
IEC 61000-4	Electromagnetic compatibility measurements and testing.

IEC 61508	Functional safety of electrical/electronic/programmable electronic safety-related system.
IEC 61850	Communication networks and systems in substations
IEC 61850-7-420	Communication networks and systems for power utility automation. Part 7-420: Basic communication structure – Distributed energy resources logical nodes
IEC 60870-5-101	Tele-control equipment and systems - Part 5 101: Transmission protocols; Companion standard for basic Tele-control tasks.
IEC 60870-5-103	Telecontrol equipment and systems - Part 5-103: Transmission protocols - Companion standard for the informative interface of protection equipment.
IEC 60870-5-104	Telecontrol equipment and systems - Part 5-104: Transmission protocols Network access for IEC 60870-5-101 using standard transport profiles.
IEC62351	Power systems management and associated information exchange - Data and communications security
IEC 60255	Measuring relays and protection equipment
IEC 61869-1	Instrument transformers - General requirements)
IEC 61869-2	Instrument transformers - Additional requirements for CTs)
IEC 61869-3	Instrument transformers - Additional requirements for inductive VTs)
IEC 62271-206	HV switchgear and control gear - Voltage presence indicating systems for rated voltages above 1 kV and up to and including 52 kV.
IEC 60947-5-1	Control switches (low-voltage switching devices for control an auxiliary circuit, including contactor relays). - Part 1: General requirements
IEC 60376	Specification of technical grade sulfur hexafluoride (SF6) for use in electrical equipment.
IEC 63360	Fluids for electrotechnical application – Specification of gases alternative to SF6 to be used in electrical power equipment
IEC 60529	Degrees of protection provided by enclosures (1P Code).
ISO 2063	Metallic coatings, protection of iron and steel against corrosion- metal spraying of zinc and aluminum.

CENELEC EN50180 Bushings above 1 kV up to 52 kV

CENELEC EN50181 Plug-in type bushings above 1 kV up to 52 kV

IEC 60282-1 High-voltage fuses - Part 1: Current-limiting fuses

IEC 60086-2 Primary batteries - Part 2: Physical and electrical specifications

In case of any deviation from the listed standard, it should be indicated in the list of deviations list submitted by the tenderer.

4. GENERAL REQUIREMENTS

4.1. General

The smart Ring Main Unit should have different configurations that consisting of a combination of various modules to meet the electricity distribution company requirements. These configurations may include, but are not limited to, the following modules:

- [X] Switch Disconnector with integrated Earthing Switch Module.
- [Y1] Circuit Breaker with integrated Earthing Switch Module.
- [Y2] Switch-fuse disconnector with H.R.C fuses with Earthing switch Module.
- [M1] Metering Tariff Module, which may be offered in one of the following configurations:
    - Includes three CTs and three VTs housed in a dedicated separate cubicle.
    - Includes three VTs in a separate cubicle, with the three CTs located within Module [Y].
    - Includes three metal enclosed plug-in VTs connected directly to the main busbar (without cubicle), with the three CTs located within Module [Y].
- [M2] Auxiliary supply module, consisting of a single or double pole voltage transformer, used in case of no low voltage feeding.

Note:

- The selection of modules [X], [Y1 or Y2], and [M1 or M2] shall be made in accordance with the relevant …. EDC requirements.
- The RMU may be optionally extendable, subject to …. EDC specifications.

4.2. Environmental Conditions

Ambient temperature	-5°C to +45°C (50°C as option according to …EDC)
Average relative humidity	95 % over a period of 24 h and 90 % over a period of one month
Maximum altitude	Up to 1000 m above sea-level

4.3. General Technical Data

The required RMU should have a safe behavior and meet all requirements given below:

Rated (Maximum operating) voltage	24 kV
Nominal operating (Service) voltage	22 kV
Power frequency R.M.S. withstand voltage (1 min)	50 kV
Impulse withstand voltage (1.2/50) µs peak	125 kV.
Rated frequency	50 Hz.
Color	Light grey (Preferred)
Auxiliary voltage	24/48 (V-DC).
IP Class (Gas Tank)	IP67
Electrolytic Copper purity	≥ 99.9 %
Copper Conductivity	not less than 57 MS/ m

4.4. Design Criteria (Indoor Use)

- A 24-kV smart automated RMU for indoor use. And shall be installed inside a building/a kiosk on the ground. It consists of a feeder switch, cable compartment, and base frame.
- Each part of RMU completed with all fittings is capable of withstanding the mechanical stress in case of operation, electromagnetic force in case of short circuit. It will not open or close by gravitation, vibration, and so on.
- The RMU structure of the switchboard should consist of a heavy-duty structural steel framework of substantial sections and dimensions to carry safely the mounted items.

- The RMU structure should be from rolled steel sheets not less than 2 mm thickness before any coating or painting for indoor use or with a mechanical impact class not less than (IK 07) and should certifies by type test.
- The RMU should be of electrostatic powder with thickness of 60:80 microns everywhere.
- For GIS: The switchgear shall be metallic partitioning classification (PM), except for fuse module may be (PI)
- The switchgear loss of service continuity category is LSC2 for [x] and [y] modules.
- The load break switch, as well as earthing switch, should be operated using a handle at the front (one handle must be provided with each switchboard)
- The positions of the load break switch and earthing switch should be indicated by a movable obvious color with suitable clear current mimic position indicator gives the position of the switches and breaker at the front of the switchboard.
- The incoming and outgoing power cables shall enter the switchgear from the bottom. These cables must be XLPE insulated, rated for 18/30 KV, and utilize aluminum conductors. The maximum conductor size permitted is:
  - For all cells: Up to 3 × 400 mm2 (or as …. EDC requirement).
  - For switch-fuse disconnector cells: Up to 3 × 150 mm2. An extension cable box is permitted for larger cable sizes
- The Load Break Switches of the ring main unit should be motorized for remote operation and control without interruption of supply.
- Manual operation without removing the motor mechanism should be possible in case of failure of the motor.
- The actuator unit of the motorization kit should be 24 or 48 VDC.
- The RMU should withstand the internal arc test, as a mandatory type test, which is intended to verify the effectiveness of the design in protecting persons in case of an internal arc (IAC).
- The internal arc class of the RMU shall be IAC AFL 16 kA for 1 Sec including the cable compartment as mandatory. The ventilation should be provided with down/upward ventilation (as option according to...EDC requirements).
- The gas tank of GIS RMU should have a pressure relief.
- A suitable base for RMU shall be provided (as optional according to …EDC requirements).
- The RMU shall be designed such that the position of the different devices shall be visible to the operator on the front of the switchboard and easy to operate & prevent access to all live parts during operation without the use of the RMU special tools.
- It is not the intent to specify herein complete details of design and construction. The offered equipment shall conform to the relevant standards and be of high quality, sturdy, robust and of good

design and workmanship complete in all respects and capable to perform continuous and satisfactory operations in the actual service conditions at site and shall have sufficiently long life in service as per requirements.

- Each module shall have a facility to carry out high voltage tests and current injection tests for the cables terminated on them via T-elbow. Disconnection of cables for testing purposes is not accepted.
- The RMU should have suitable support facilities to be lifted from the top and wooden pallet to be lifted from bottom for safe transport.
- If the horizontal earthing busbar is exposed, it shall be enclosed by a sheet-metal cover, including all relevant bolts.

4.5. Switch Disconnector (Load Break Switch)

- Switch disconnector shall be gas load break switch, 24 kV - 630A, manual drive and motorized with its spring mechanism, used for the cable feeders.
- The interruption medium can be either gas insulated or vacuum. The gas shall be in compliance with IEC 60376 or IEC 63360.
- Switch disconnector shall have three positions; 'ON', 'OFF', and 'Earth'. The main switch and earth switch should have two independent manual operating mechanisms to prevent simultaneous closing of the main switch and earth switch.
- The manual switching operation shall be using an operating handle. Switching speed shall be independent of the handle speed.
- The electrical local, remote, and tele-controlled switching operation shall be done using the geared motor mechanism with associated closing and opening coils, interposing relays, and so on.
- The switch disconnector shall have the retrofitting capability on the site for the operating mechanism and associated equipment.
- Switch disconnector shall be designed for interrupting the full rated current, interrupting small inductive current for unloaded transformer or capacitive current for cable or overhead lines load breaking, and fault-making.
- The earth switch contacts shall be designed to close into a fault and shall have the same short circuit capacity as the main contacts.
- Necessary auxiliary contacts for controlling, signaling& free contacts (2NO-2NC) at least in low voltage compartment and all accessories needed for operation should be provided for each of the switch disconnector and earth switch.
- The operating handle shall be designed to prevent the opening of the switch-disconnector or the earthing switch after closing

Switch disconnector	**Rated Voltage**	24 kV
	Rated normal current at 40 C°	630 A
	Short-time withstand current (I_K)	20 kA for 1 second
	Making capacity	50 kA
	Mechanical endurance	M1: (1000 operating cycles)
	Electrical endurance	E3, C1
Earth switch	**Short-time withstand current**	20 kA for 1 S
	Making capacity	50 kA
	Mechanical endurance	M0: (1000 operating cycles)
	Electrical endurance	E2

4.6. Switch-Fuse Disconnector

- The LBS shall be gas insulated - load break switch, 24 kV - 200A, manual drive and motorized with its spring mechanism.
- The interruption medium can be either gas insulated or vacuum. The gas shall be in compliance with IEC 60376 or IEC 63360.
- It used for the transformer protection and complete with three high rupture capacity (H.R.C) fuses and the rating of fuse should be according to transformer rating and should be specified by.…EDC on every tender.
- The high rupture capacity (H.R.C) fuses shall be designed according to the latest revision of EDMS 19-400.
- The LBS shall have a 3-position ‘ON’, ‘OFF’, and ‘Earth’ the main switch, and the earth switch should have two independent manual operating mechanisms to prevent simultaneous closing of the main switch and earth switch.

- The manual switching operation shall be using an operating handle. Switching speed shall be independent of the handle speed.
- The electrical local, remote, and tele-controlled switching operation shall be done by mean of geared motor mechanism with associated closing and opening coils, interposing relays, and so on.
- The switch fuse disconnector shall have the retrofitting capability on the site for the operating mechanism and associated equipment.
- The operating handle shall be designed to prevent the opening of the switch-disconnector after closing.
- The fuse holder frames should fit with an automatic opening device that functions even if only a single fuse blow.
- Necessary auxiliary contacts for controlling, signaling& free contacts (2NO-2NC) at least in low voltage compartment and all accessories needed for operation should be provided for each of the switch disconnector and earth switch.
- The fuse striker mechanism should trip the switch in case of any failure (The automatic fuse contact shall be designed to open into a fault) and shall have visual and signal indications.
- The switch shall have the capability to open electrically by Buchholz relay via trip coil.

Switch fuse disconnector	**Rated Voltage**	24 kV
	Rated normal current at 40 C°	200 A
	Short-time withstand current (I_K)	20 kA for 1 second
	Making capacity	50 kA
	Mechanical endurance	M1:(1000 operating cycles)
Earth switch	**Short-time withstand current**	20 kA for 1 S
	Making capacity	50 kA
	Mechanical endurance	M0: (1000 operating cycles)
	Electrical endurance	E2

4.7. Circuit Breaker

- The Circuit breaker shall be of fixed type.
- The interruption medium can be either gas insulated or vacuum.
- The circuit breaker shall have three positions; ‘ON’, ‘OFF’, and ‘Earth’ or the circuit breaker shall have two positions; ‘ON’, ‘OFF’, and should be connected with disconnector switch to be able to safe isolation for feeder, and the feeder can be earthed.
- The circuit breaker and earth switch should have two independent manual operating mechanisms to prevent simultaneous closing of the circuit breaker and earth switch.
- The circuit breaker, disconnector, and earth switch should have independent manual operating mechanisms to prevent simultaneous closing of the circuit breaker, disconnector, and earth switch in case of having isolator in the CB compartment.
- The manual switching operation shall be using an operating handle or push buttons. Switching speed shall be independent of the handle speed.
- The circuit breaker shall have anti-pumping and anti-slam protection.
- The opening process of the circuit breaker shall be mechanically by trip button, and electrically by protection relay, by Buchholz relay via trip coil, and local/remote signal (24/48 VDC).
- The closing process shall charge the operating mechanism for the opening process
- The electrical local, remote, and tele-controlled switching operation shall be done using geared motor mechanism with associated closing and opening coils, interposing relays, and so on.
- The circuit breaker shall have the retrofitting capability on the site for the operating mechanism and associated equipment.
- The circuit breakers should be mechanically and electrically trip free and with provision for manual operation.
- Necessary auxiliary contacts for controlling, signaling& free contacts (2NO-2NC) at least in low voltage compartment and all accessories needed for operation should be provided for each of the switch disconnector and earth switch.
- The operating handle shall be designed to prevent the opening of the switch-disconnector or the earthing switch after closing.

Circuit Breaker	**Rated Voltage**	24 kV
	Rated normal current at 40 C°	200/400/630 A - As required
	Short-time withstand current (I_K)	20 kA for 1 second
	Short circuit breaking capacity (I_{SC})	20 kA
	Making capacity	50 kA
	Mechanical endurance	M1: (2000 operating cycles)
	Electrical endurance	E2, S1, C1
	Opening time	Up to 75 ms.
Earth switch	**Short-time withstand current**	20 kA for 1 S
	Making capacity	50 kA
	Mechanical endurance	M0: (1000 operating cycles)
	Electrical endurance	E2

4.8. Instrumental Transformers/Sensors

4.8.1. For smart metering tariff Module [M1]

- Metering Module may have different configurations:
  - Includes three CTs and three VTs housed in a dedicated separate cubicle.
  - Includes three VTs in a separate cubicle, with the three CTs located within Module [Y].
  - Includes three metal enclosed plug-in VTs connected directly to the main busbar (without cubicle), with the three CTs located within Module [Y].
- Three single phase dry type (conventional/ring) current transformers according to EDMS17-400 specification.
- The block type CT burden should be 5VA class 0.5S.
- The Current transformer shall comply with IEC 61869.
- The current transformers should be interchangeable from lower ratio to higher ratio from the secondary side of the current transformer only.

- The current transformers ratio should by specified by….EDC on every tender.
- The winding of the current transformers should be made of copper.
- Suitable current transformer shall be a part of RMU compartment along with its complete sizing calculations feed from bus-bar or outgoing feeder.
- It is permissible to utilize the block-type CT from the [M1] module for protection purposes, provided that the [M1] module exclusively supplies the [Y1] module.
- For ring CT:
  - Highest voltage for equipment Um (R.M.S) 0.72 kV
  - Rated power frequency withstands voltage 3 kV
- Three single phase (conventional/screened) voltage transformers dry type according to EDMS17-401.
- The VT shall comply with IEC 61869.
- The voltage transformer shall be (22000/√3/110/√3 V/110/3 V) 50 VA, CL0.5 Voltage factor 1.9 for 8 hrs. & 1.2 continuous feed from bus-bar/outgoing feeder.
- Conventional voltage transformers should be protected by suitable H.R.C fuses on the Primary side. No need for fuses in case of using screened VTs without life parts in the compartment.
- VTs shall be protected by suitable MCBs on the secondary side. Suitable protection system shall be provided to protect the voltage transformer against Ferro-resonance phenomena.
- For conventional CTs and VTs, the leakage path should not be less than 2 cm/kV of the nominal operating voltage.
- Separate box of IP 42 for installing smart energy meter and its connections. The meter dimension (L×W×D) is not less than (293×178×95) mm. Door-open signals for both meter box, VT and CT compartment shall be wired to the terminal block inside the meter box. The front door of the meter box shall be transparent.

4.8.2. <u>For auxiliary supply module [M2] in case of no low voltage feeding</u>

- Single / double pole (conventional/screened) voltage transformers dry type according to EDMS17-401.
- The voltage transformer shall be (22000/√3/suitable value), voltage factor 1.9 for 8 hrs. & 1.2 continuous feed from bus-bar.
- The secondary voltage and burden of the voltage transformer should be design according to the manufacture solution and the RMU configuration.
- The rated continuous output power shall be design regarding to the consumption with safety

factor not less than 20%.

- Conventional voltage transformers should be protected by suitable H.R.C fuses on the Primary side. No need for fuses in case of using screened or double pole VTs without life parts in the compartment.
- VTs shall be protected by suitable MCBs on the secondary side. Suitable protection system shall be provided to protect the voltage transformer against Ferro-resonance phenomena if applicable.
- For conventional CTs and VTs, the leakage path should not be less than 2 cm/kV of the nominal operating voltage.

4.8.3. For feeders and transformer Modules [x] and [Y]

- The three single phase current transformers may be Block type CTs, or Ring core CTs, or Current sensors.
    - The Current transformers (Block type or Ring core) shall comply with IEC61869.
    - The Current Sensor shall comply with IEC 60044-8.
    - The current transformers ratio should be according to rated normal current of each module.
    - The winding of the current transformers should be made of copper.
    - The current transformer or current sensor used for protection shall provide both measurement and power output for the protection relay.
    - The ring core current transformer (CT) for protection relays shall be of accuracy class 5P10 for a ratio of (…/1 A), or class 5P20 for a ratio of (…/5 A). The rated burden of the CT shall be more than the total connected burden such as protection relay and lead wiring.
    - The Ring core or the current sensor used for measuring which connected to the RTU or to the FPI shall have class 1 for measurement.
    - The block type CT class 5P20 for protection relays (for [Y1] module). The rated burden of the CT shall be more than the total connected burden such as protection relay and lead wiring.
    - The current sensor/ Ring core CT should be designed to give a thermal protection to the measuring and protection devices during heavy earth faults and should allow for cable fault currents up to 20kA (rms) and should operate at cable sheath temperature up to 90ºC.
    - Suitable current sensor shall be a part of RMU compartment along with its complete sizing calculations.

- The current sensors/ Ring core CT should have current injection facility to perform the primary current injection test (preferred).
- The current sensors/ Ring core CT shall has rated and insulation voltages of 0.72 kV/3 kV - 1min.

- Voltage Detecting and Indicating System (VDIS) should be integrated in all functional modules.
  - VDIS shall receive the signals through the capacitive sensor installed inside the bushings of RMU which detect the cable side voltage.
  - The indicator shall be flush-mountable.
  - VDIS shall be according to standard IEC 62271- 213.
  - VDIS shall have a low voltage hot phasing facility. The lamps / LCD shall be powered by bushing-type capacitive dividers.
  - VDIS shall detect the voltage presence /absence and be fitted with a 3 LED/LCD.
  - VDIS should deliver the measured signal to the RTU or to FPI.

4.9. Interlocks

- Interlocks between different components of the equipment are provided for reasons of protection and for convenience of operation. Interlocks shall not be damaged by attempted incorrect operations of any associated switching device.
- The interlock shall be provided between electrical and manual operations.
- A mechanical indicator should be provided to show the position of the switches to enable locking at each position separately.
- Manual closing blocking facility for a gas circuit breaker in AIS RMU.
- Electrical closing or opening inhibition of LBS or CB in case of low gas pressure.
- Electrical closing inhibition of LBS or CB in case of cable door is open.
- Arrangements for padlocking should be provided for:
  - Each position of the switches enables locking at each position separately.
  - Electrical Pushbuttons of closing and opening the switches.
  - Mechanical pushbuttons for closing or opening the circuit breaker.
- The switch should be provided with the inter-locks specified in IEC62271-200 but not limited to the following:
  - Operation of the ‘ON/OFF’ mechanism of switch / CB shall be possible only if the earth switch is in the ‘Earth OFF’ position.

- Operation of the 'ON / OFF' mechanism of the earth switch shall be possible only if the mechanism of switch / CB is in the 'OFF' position.
- In case of using disconnector in series with the CB:
  - Operation of the „ON/OFF‟ mechanism of disconnector shall be possible only if the earth switch/ CB are in the 'OFF' position.
  - Operation of the „ON / OFF‟ mechanism of the earth switch shall be possible only if the mechanism of disconnector is in the „OFF‟ position.
- The cable box cover cannot be opened unless the earth switch is in the "ON" position.
- LBS or CB cannot be moved "ON" position when the cable box cover is in the open position.
- Earth switch cannot be closed unless the cable is de-energized (as option according to …EDC requirements)
- Fuse changing or accessing the fuse chamber cannot be done in any position other than the "Earth" position.

- Additional lock for fuse module:

Fuse Switch cannot be moved "ON" position when:

- The fuse access cover is not properly closed
- Fuse switch in 'Earth' position.

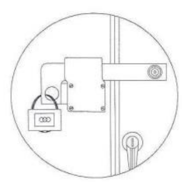
4.10. Gas pressure Indicator & Refilling provision

- Temperature independent gas pressure gauge shall be provided. The safe operating zone shall correspond to a temperature range of –10 °C to 50 °C.
- A refilling non-return inlet valve if provided shall be easily accessible for field refilling.
- The vendor shall give a guarantee for the tank for maximum leakage rate of gas will be lower than 0.1%/ year.
- A gas gauge shall be provided for visual indication of gas pressure inside the switchgear chamber. The gas gauge shall be readily visible from the front of the unit without the necessity to remove any covers and be clear marked to indicate the normal gas pressure and low gas pressure on the gauge face.

4.11. Enclosure (Outdoor Use):

- Outdoor RMU shall have a tamperproof and a weatherproof steel enclosure which cover whole body of the RMU and control box.
- The Enclosure should be of electrostatic powder with thickness of 60:80 microns

- The status of the door of the enclosure shall be monitored.
- The enclosure shall provide with lockable doors, door handles, ventilation louvers and lifting hooks.
- The doors of enclosure should be same design of M.V compartment of kiosk.
- The degree of protection shall be IP54.
- All nuts, bolts and washers shall be hot dipped galvanized.
- The enclosure shall have a pocket or provision inside to store the instruction documents and other relevant information.
- The enclosure should be from rolled steel sheets of 2.5 mm thickness before paint at least or with a mechanical impact class not less than IK10 for outdoor use and should support by certificate.
- The roof of enclosure should be galvanized steel, and finished with electrostatic paint.
- The roof of enclosure should have slope (towards to sides) by degree not less than 6°.
- The enclosure should be hot-dip galvanized (HDG) to provide distinct corrosion protection, and finished with electrostatic paint (as option according to ….EDC requirements).
- Natural ventilation shall be provided.
- The enclosure door should be provided with suitable gaskets, and should have locks with a master key.
- The enclosure shall be equipped with a base of iron beam with supports to withstand lifting it from the bottom and suitable for the RMU weight. The iron beam shall be painted by two layers one for iron erosion resistant and the 2nd is black.
- Two large galvanized iron bolts (approved closing means) shall be installed, one on the top and the other on the bottom of the left leaf for easy closing and opening the door.
- The doors shall be equipped with two eye-bolts (iron hasp) as shown in the figure. In addition, one eye-bolt to be supported by locks.



- The means for non-closing the external doors during maintenance is by mean of a rectangular section attached to the body of the panel with a screw at one end and with a hole on the other end that is fixed with a stent welded to the outer door when it is opened.

4.12. Fault passage Indicator (FPI) for cable feeders:

- The FPI may be used as integrated function into the measuring unit of each feeder.
- The FPI is required for each feeder, mainly indicate the following features:
  - Directional Phase to phase fault.
  - Directional Phase to earth fault.
- The FPI fault shall be feed from the VDIS and current sensors or ring core CT.
- The FPI device should sense the direction passage of the fault current through the cable.
- FPI should be equipped with self-supervision facility.
- FPI should have the flexibility to reset by different ways such as the healthy starting transient current, the presence of the voltage, Fault current is cleared, adjustment time delay, and software.
- FPI shall integrate with external outdoor fault lamp indication LED type (fixed on visible place) should emit a blinking red light with IP54 according to IEC 529, lamp auto reset at healthy condition. Connection of this outdoor unit is not compulsory.
- In case of separate FPI the following feature shall be included:
  - Compatible with RTU communication protocol, support Modbus TCP or Modbus RTU or 60870-101/104 or IEC61850 communication protocols.
  - Support enough NO/NC output contacts operated with fault condition for covering all required signals in case of separate module.
  - Protection of enclose IP 40 or above.
- FPI shall have a latched output when the current exceeds the threshold value for overcurrent and the earth fault.
- The reset to the healthy condition should include: (Fault current is cleared, adjustment time delay, software, sense current due to the cable re-energize, and sense the voltage due to the cable re-energize).
- FPI shall have a time-tagged events
- Easily replaced and maintained.
- Compact size.
- Withstand ambient temperature range of 0 up to +50°.
- Not responsive to mechanical shocks.
- Protected against sun, radiation, humidity and dust.
- It should be provided with an indication for transient and permanent faults.
- Measuring load current for each phase.
- Phase to earth fault function:
  - Indicator should support adjustable setting range (10-180 A) primary current with suitable step.

    - Indicator should support fast act time response at fault condition (0.06 Sec. or less).
- Phase to phase fault function
    - Indicator should support wide adjustable setting range (225-700) primary current with suitable step.
    - Indicator should support fast act time response at fault condition (0.06 Sec. or less).

4.13. Capacitor divider and phase comparator

- Every load break switch or circuit breaker cell should be equipped with three capacitor dividers connected to the voltage presence indicator system monitored by LEDs or LCD to indicate the voltage presence on the cable.
- There should be a facility to check the phases concordances with the use of external device. This external device shall be provided with the RMU.

4.14. Protection Relay for [Y1] module

The protection relay shall be three phases combined non-directional (over current + earth fault & short circuit) digital protective relay and timer in one unit connected to three current transformers/sensors of the protected object and suitable for resistance earthed neutral system, in solidly grounded system the residual over current protection can be used on feeder of any length and fulfilling the following requirements and specifications:

4.14.1. Technical Specification

Protective relay should select based on the following specifications:

- The protection relay is based on a multiprocessor environment.
- The modern technology should be applied in both hardware and software solution.
- The relay must be designed for protection against (O.C. & E.F & S.C & Thermal) includes the following:
- Reset factors > 95%
- Operation temp. From -5oC to +55oC
- Humidity > 90% non-condensation.
- It should be able to operate without intended delay.
- Definite and inverse time characteristics according to IEC standards.

- Two LEDs (for ready of protection trip indications) with suitable reset pushbutton and can be automatically rested and include one LED for each fault type and monitoring.
- Self-monitoring and blocking for internal faults with alarm contact and led indication.
- Support a digital screen for current measurements.
- The relay shall have external trip capability to trip the CB if Buchholz relay trip in case of oil transformer or temperature high for Dry transformer by mean of digital input.
- The three phases overcurrent unit and the earth fault measure the phase current and the neutral current of the protected object.
- The relay should be flush mounted.
- The relay must be digital type and combatable with RTU communication protocol which supports Modbus or IEC-61850 or any communication protocol.
- Output contacts (Tripping contacts).
  - Rated voltage: related to the auxiliary supply voltage with ripple content (3-4) %.
  - Continuous carrying capacity 2 ADC.
  - Making and carrying current capacity for 0.5 sec > 6 A.
  - Make and carry for 3.0 sec > 4 A
  - Free contacts at least (1 N.O for O.C & 1 N.O for E.F) besides the watch dog contact.
- The actuator shall be a low power tripping coil specifically designed to operate with the sensors and the processing unit with minimum energy.
- Three-phase thermal protection.
- The relay must have the inrush current blocking based on the second order harmonic percentage which can be used in all protection stages, without using multiplayer
- Support a digital screen for current measurements, parameter, setting and events.
- All parameters and events can be set or read by P.C or manually by key pad
- Smaller dimensions preferred
- Tender must supply with offer detailed instruction and catalogue
- The relay must include event recorder and should be read by screen or software.
- Continuous self-supervision and diagnosis of electronics and software
- User-selectable password protection for HMI
- Display of primary/secondary current values.
- Housing degree of protection IP 54 according to IEC 60529.
- The output contact functions freely configurable for desired operation.
- The relay should have at least two setting tables.
- Not responsive to mechanical shocks.
- Protected against sun, radiation, humidity and dust.

- RMU manufacturer to provide suitable supplying for protective relay must have a dual power:
  - Self-powered: energized by the CT's
  - External supply: energized by uninterrupted power supply in case of communication required.
- Complied with MV relay standard IEC-60255.
- The protection relay shall be approved form the protection central sector of EEHC.

4.14.2. <u>Non-volatile memory for the following:</u>

- The digital protective relay should be equipped with a nonvolatile memory for preserving important data during auxiliary supply breaks. The memory does not need batteries, and a lifelong service is guaranteed, and following data is stored:
  - Date and Time.
  - Setting values
  - Recorded last five fault data.

4.14.3. <u>The real-time clock</u>

- The relay must be supported by a real-time clock (RTC) which used for time stamping of events. It is also running during auxiliary power breaks. When the supply is RE-established, the relay sets the right time and new events are stamped accordingly.

4.14.4. <u>The relay must include the following protection:</u>

- I > Three-phase low-set over current stage with definite–time and inverse definite minimum time characteristic (IDMT)
- I >> Three-phase high-set over current stage with the instantaneous or definite time characteristic.
- Io > low-set non-directional earth –fault stage with definite-time or inverse definite minimum time characteristic (IDMT).
- Io >> High-set non-directional earth-fault stage with instantaneous or definite –time characteristic.
- Setting values and ranges:
  - Current stages (over current)
    - Low set over current I > (0.1 – 2.5) In with steps < 0.05 In.
    - High set over current I >> (0.2 - 20) In with steps < 0.05 In
  - Operating time ranges (over current)
    - Operate time in DT mode: (0.05 - 3) sec with step < 0.01 sec
  - Current stages (earth fault)

- Low set earth fault Io>: (0.1 – 1) In with steps < 0.01 In.
- High set earth fault Io>>: (0.2 - 10) In with steps < 0.01 In.
- Sensitive earth fault: (0.005 – 0.5) In (as option by a core balance CT)

- Operating time ranges (earth fault)
  - Operate time in DT mode: (0.05 - 3) sec with step < 0.01 sec.
  - Sensitive earth fault: (0 – 3) sec. (as option by a core balance CT)

4.14.5. Standard tests

- **Insulation tests**
  - Dielectric tests: according to the (IEC 60255/5 &60225/5 or equivalent) tests voltage 2 KV 50 Hz, 60 sec.
  - Impulse voltage test according to the (IEC 60255/5 & 60225/5 or equivalent) Tests voltage 5 kV, unipolar impulses, waveform 1.2/50µs.
  - Insulation resistance measurements according to the IEC60255/5.

- **High frequency interference withstands:**
  - According to class III of IEC 60225
- **Over load**
  - Current circuit temporary $\geq$ 25 kA primary for 1.0 sec.
- **Mechanical tests**
  - Vibration test (sinusoidal) according to the IEC 60225.
  - Shock and bump test according to the IEC 60225.

4.14.6. Diagnostic and Supervision

- Supervision functions for energizing current input circuit.
- Self-test diagnostic.

4.14.7. Measurement

- Display primary current values (IR-IS-IT-IN).
- Transient disturbance Events.

4.14.8. Communications:

The relay should support local and remote communication, local communication with P.C or laptop if applicable and suitable remote communication to be connected with the RTU.

4.14.9. Associated current transformer

Suitable current transformer for protective relay shall be a part of RMU compartment along with its complete sizing calculations. It shall be for measurement and powering purposes.

4.15. General requirements for GIS

4.15.1. General

- The gas shall be in compliance with IEC 60376 or IEC 63360.
- Busbar shall be copper or aluminum and housed in a sealed for life gas tank.
- Protection degree IP:
- L.V mechanism compartment / cable compartment: IP2X/IP2X respectively. It is necessary to close and cover all openings with the panel front well so as not to allow any dust penetration into the panel or the mechanism.
- All load break switches should be accommodated in one tank and may be accommodated every load break switch to be on a separate tank (as option according to …. EDC requirement).
- The gas pressure indication for the tank should be through two stages (Low for alarming– insufficient for operation blocking) or one stage in case of completely sealed tank.
- The envelope should be permanently sealed and filled with gas at suitable pressure.

4.15.2. Terminal bushing/Cable Box

- Terminal bushing in the RMU shall be suitable for cable termination inside cable box suitable for acceptance single/three core AL/CU XLPE insulated cables up to 400/630 mm2.
- Each cable box shall have a bottom plate if applicable and a cable clamp. Bottom plate shall be in two halves with cable entry hole of suitable diameter. Cable clamp shall be detachable semi-circular halves suitable to hold the cable inside the cable box without cable glands. Suitable rubber fitted to each cable entry hole.
- Terminal bushings for RMU shall be interface type C with M16 bolted contact for terminating cables with the use of L elbow type or T elbow type.
- Vertical distance from the top of cable clamp to centerline of cable bushings shall be suitable for all type of terminations
- Removing and installing of cable box cover shall be with minimum number of bolts.

4.15.3. Earthing

An earthing bar made of copper shall be provided and bolted to the frame. It shall be positioned to facilitate the earthing of cable sheaths, elbows, and earthing devices. The cross-sectional area of the earthing bar shall be the same as that used during the type test.

4.16. General requirements for AIS

- The gas pressure indication for the insulating gas (LBS and CB) should be through two stages (Low for alarming– insufficient for operation blocking), or one stage in case of completely sealed tank.
- Heater of 100-watt, 220v AC with a hygrostat and a M.C.B per each module in cable compartment.
- Additional 100 watt, 220v AC heater shall be installed in the bus bar compartment with its associated MCB and hygrostat.
- Protection degree IP3X.
- The creepage distance of the resin insulator should be not less than 2 cm/KV.
- Main Bus Bar:
  - It should be insulated B.B system and made of high grade of electrolytic copper at least 630A for all RMU panel should not be less than 400 mm^2.
  - The temperature rise should be according to IEC 62271-200.
  - All main bus bars should be insulated by red anti-track heat shrinkable tubes and the connection points should be covered with a suitable insulated cover.
  - The bus-bar compartments should be placed at the upper part of the switchboard and fitted with using dielectric supports.
  - The entire bus and structure should be constructed to withstand the short circuit effects due to the rated short circuit current of 20 kA for 1 Sec at least.
  - The Bus bar connections should be anti- oxide greased.
  - Main B.B should be extendable and securely mechanically bonded to each unit.
- Earth Bus Bar:
  - An earth copper bar at the bottom of the switchboard extending the entire length of each distribution panel should be provided with a cross section area to withstand the rated short circuit current of the system the bus should be extendable and securely mechanically bonded to each unit.
  - An earthing bar made of copper shall be provided and bolted to the frame. It shall be positioned to facilitate the earthing of cable sheaths, elbows, and earthing devices. The cross-sectional area of the earthing bar shall be the same as that used during the type test.

5. RMU Automation

The RMU must contain an integrated automation solution to be able to control and monitor it remotely via the SCADA.
The automated RMU shall provide the ADMS by the required data to be able to carry out the required functions.

5.1. Low Voltage Box

- It should be stand on the top of the RMU with suitable clearance and shall be accessible from the front.
- It shall comply with IEC 529, with an index of protection IP 42.
- The enclosure shall provide housing for the RTU hardware, battery / PS, modem MCBs and auxiliary relays.
- The enclosure should be Pad lockable and fabricated using 2 mm thick.
- The low voltage panel door shall be provided with key lock (master key)
- Suitable air cooling (forced or natural) is required as per manufacture design.
- Operating at ambient temperature range from 0C° to + 50 C°.
- All wiring in the RMU shall be labeled directional from / to.
- Each control/protection circuit shall be protected by appropriate MCB.

5.2. Power Supply (PS)

- The PS shall provide the power for the RMU automation devices besides charging the battery.
- The PS shall operate the motor mechanism with/without the battery.
- The input voltage of PS:
  - Shall be 220 V / 50 HZ with voltage fluctuation within ± 15 %.
  - Shall be suitable with the secondary voltage of the voltage transformer in case of using [M2] module only.
- The PS shall be protected by suitable MCB according to its capacity.
- The PS output voltage and capacity shall size and designed to supply the motors mechanisms, coils, communication devices, indications, auxiliary relays, RTU, protection relays, and FPIs)
- The capacity of the PS shall be design by the manufacturer, 20 % safety factor shall be considered.
- The PS shall be temperature-compensated charger

- The PS shall have a protection against over and under voltage for the input and output sides.
- The PS shall have a protection against overcurrent and the short circuit.
- The output DC voltage shall be constant with ripple voltage less than 1 % without battery.
- The PS shall test and monitor the battery availability, status, capacity indications.
- The PS shall have a charging current limitation for battery.
- The PS shall have a protection against deep discharging for the battery.
- The PS shall have a protection against overcharging of the battery.
- The battery shall back up the power for the RMU automation devices during the outage of the auxiliary power supply.
- The backup can be by two batteries on for RMU operation and the RTU and the 2nd for RMU automation devices. (as option as per …EDC requirements
- The battery shall be sealed type with maintenance free.
- The battery sizing shall be designed by the manufacturer to back up the supply for the RMU devices for (5 to 8) hours (as option as per …EDC requirements) and perform at least 10 operation cycles.
- The battery shall be long life industrial usage 10 years classifications.
- The PS and battery shall be suitable to operate at ambient temperature 50 C°.
- The signal list of the PS and battery are illustrated in the signal list.
- Insulation:
    - Dielectric Rigidity and Voltage Surge withstand.
    - The equipment must be able to support between all the terminals connected together and the mass, as well as between each of the circuits galvanically independent and all the others connected together and with the mass:
        - 2 kV rms at 50 Hz, 5 kV peak, in voltage shock (standardized lightning shock 1.2/50µsec).
        - Dielectric rigidity between open contacts of the output relay must be 1 kV rms at 50 Hz.
    - The LV alternating power supply must be isolated from the other circuits by a transformer offering the following rigidity between phase and mass:
        - 10 kV rms at 50 Hz, 20 kV peak, in voltage shock (standardized lightning shock 1.2/50µsec).

5.3. Industrial Router

5.3.1. General Features:

- The router shall be an industrial-grade 3G/4G model, supporting LTE on frequency bands B1, B2, B8, B20, and B28.
- Fan less design with passive (natural) heat dissipation.
- Operating temperature range: –40 to 60°C standard operation, –40 to 75°C in a forced air enclosure, suitable for harsh industrial environments.
- The RTU shall communicate with a modem (router) or two modems (as option as per …EDC requirements) with routing features in order to communicate wireless 3G/4G or fiber optics.
- The modem shall have two SIM cards when a single modem is used and a single SIM card when two modems are deployed.
- Capable of controlling the RSSI (Received Signal Strength Indicator) threshold for automatic SIM switching.
- Operates on a wide range of DC input voltage.
- Capable of connecting to external 4G outdoor antennas for improved signal reception.

5.3.2. Interface Ports:

- Gigabit Ethernet (GE) electrical ports.
- Integrated RS485 / RS232 serial communication interfaces.

5.3.3. Event Notification and Monitoring:

- SNMP (Simple Network Management Protocol) support for remote monitoring.
- System log generation for event tracking and diagnostics.
- Fully manageable via a centralized Network Management System (NMS).

5.3.4. Networking and Security Capabilities:

- The router shall support standard static and dynamic routing protocols.
- Comprehensive Virtual Private Network (VPN) functionalities shall be provided, including IPsec tunneling.
- Advanced security features including IPsec-based encryption, and user authentication via AAA (Authentication, Authorization, and Accounting) and RADIUS protocols

5.3.5. Integration Requirement:

- The supplier is responsible for ensuring seamless and secure integration of the router with the existing Distribution Control Center infrastructure.

5.4. Remote Terminal Unit (RTU)

5.4.1. RTU Specification & Architecture

- This specification includes the requirements for remote terminal unit for acquisition of real time status and control functions associated with selected 24 KV RMUs.
- The RTU shall provide remote control, monitoring and data acquisition function and interface to the SCADA system via communication device.
- EDC reserves the right to require a documented demonstration of the RMU/RTU capabilities to support DMS SCADA functionality.
- The RTUs should have an architecture that supports convenient installation, maintenance and expansion features.
- The RTU should include all hardware, software and firmware necessary to meet the I/O point requirements including input and output cards and output relays.
- The RTUs configuration should include a central processing unit module, I/O module digital and analogue type.
- The RTU shall have a Built-in webserver for commissioning, upload and download the configuration program and maintenance with local and remote access.
- The RTU shall certify with IEC 61000-4.

5.4.2. Environmental Requirements

- RTU and its modules should be capable of operating at ambient temperature up to 50 C°.
- RTU should be capable of withstanding a relative humidity 95% non-condensing.

5.4.3. RTU Functions

- The RTU shall have the basic functions including remote control, measurement, detecting event/alarm, recording, information points from devices, relays and/or IED's data communication, remote/local switching, internal clock, time synchronization, time stamp/tag, data holding, diagnosis of itself in failures and etc.
- RTU shall be multi- center multi-protocol.
- Receiving and processing digital and analog control commands from the master station or SCADA.

- RTU shall support Sequence of Events feature with 1 msec resolution with time stamping and buffering capacity of SOE storage shall be at least 1000 events and to be accessed from the ADMS and at site.
- RTU shall have logic functions to allow customized automation schemes and should be ready to activate an IEC61131-3 logic engine or similar.
- Interface shall be preferably web-based.
- It shall be possible to implement the alarming feature.
- The RTU data/time shall not affected by the loss of the power supply.
- RTU shall act as a data concentrator for acquiring data from slave RTU's and exercising supervisory control on slave RTU's.
- RTU shall accept polling messages from at least two master stations simultaneously using separate logical databases for each master station.
- RTU shall communicate simultaneously on all communication ports.
- The RTU shall have the option of redundant communication through the Ethernet ports.
- It shall be possible to export database to excel or similar software application.
- Data transmission rates up to 9600 baud for serial ports and 10/100 Mbps for TCP/IP.
- RTU shall have the capability of automatic re-start after a power outage.
- RTU shall be scalable.
- The RTU shall be designed to communicate with a minimum 10 Electronic devices or IED's connected with 100 variables per device.
- The RTU shall be capable to calculate all logical functions needed by an Advanced Distribution Management System such as Power factor, Active power, Reactive power.
- RTU and Control box shall be illuminated by a lamp for easy O&M.
- The RTU capable to interface with the programming laptop (Programming, troubleshooting, etc.).
- The software should be providing automatic restart of the RTU upon power restoration, memory parity errors, hardware failures, and manual request.
- The software should initialize the RTU and being execution of the RTU function without intervention by the SCADA/DMS master station.
- It shall be possible to allow access from a remote location (e.g from the control center) to the IEDs connected to the RTU in the RMU while keeping the RMU LAN and Control Center LAN separated from each other.
- All requests should be reported to the SCADA/DMS.
- In order to easily support the system under continuously changing site conditions all protocol, configuration, and application data must be contained in easily programmable non-volatile

memory such as flash EPROM and the protocol which is adopted to RTU shall be configured through ADMS and / or by using maintenance tool.

- The RTU design should be independent of any communication protocol that would impose restrictions on the flexibility of functionality of the RTU.
- Protocol changes should be accomplished by software/firmware changes only.
- The RTU shall have two temperature sensor inputs compatible with PT100 to measure the transformer temperature in case of delivering [Y] module.
- It shall be possible to retrieve the configuration of an existing RTU without the need of the original configuration backup.
- The RMU unit shall be provided with a separate Local/Remote switch inside the LV cabinet and may be on the LV cabinet (as option as per …EDC requirements).
    - The switch shall be changeover switches and covered for prevention of unintentional changeover and can be operated from the site only.
    - Each switch shall have double-status. Signal of both statuses shall be sent to the ADMS.
    - The switch defines behavior of the RTU as follows;
        - 1) In Remote position: the DMS can control LBS remotely. Open/Close command is accepted by the RTU, and RTU outputs a control signal to the target LBS. Manual operation of LBS at site is not accepted. After switching to Local position at site, the manual operation can be carried out.
        - 2) In Local position: Open/Close LBS can be done in manual at site. The control command from DMS is not accepted by the RTU.

- Lock/Unlock (for each cell), which can control each cell individually (as option as per …. EDC requirements)

5.4.4. Central Processing Unit Module (CPU)

- Central processing unit (CPU) should handle all protocol emulation i.e. IEC60870-5-104, MODBUS master …etc., perform data acquisition, and control requests.
- The CPU should accept commands from the master station, perform address recognition, assemble response message in accordance with the received command message, and transmit theses message to the SCADA / DMS master station.
- The CPU should have RTC and provide interfaces for time synchronization from DCC over IEC60870-5-104.
- The CPU should have user configurable routines/procedures to carry out connection establishment, link failure detections and reconnection after failures for dialup connectivity.

- THE CPU should manage communication between all other function modules of RTU and should determine the integrity of the RTU.
- The processor should provide diagnostic information in the message structure that the SCADA / DMS should monitor.
- A flag should be set if the RTU perform a restart for any reason including power failure.
- The CPU should be programmable in a high language.
- REL should be able to program the RTU and manage the RTU database from the RTU test set and download parameters and configuration data from the SCADA / DMS system.

5.4.5. Communications

- The RTU shall support different protocols to communicate with the SCADA/ADMS as a server/slave:
    - IEC 60870-5-104 as mandatory
    - and/or IEC 61850, IEC 60870-5-101/104, DNP 3
- The RTU shall support different protocols to communicate with the RMU automation devices and protection relay as a client/Master:
    - IEC 61850
    - and/or IEC 60870-5-101/103/104, DNP 3, Modbus
- The RTU shall support different interfaces of communications:
    - Two Ethernet ports
    - At least one serial port RS 485.

5.4.6. I/O MODULE

- Each I/O module should be capable of interfacing with digital inputs, control output points and combination of point's types.
- I/O modules should be replaced without reprogramming, redefinition of configuration parameters or rewiring.
- The RTU should include facilities for handling status input and control output points. Requirements for each type of I/O point.
- A control disable switch should be provided within each I/O module.
    - When the switch is in the control position, the SCADA/DMS or test set should have control of the digital control outputs.
    - When the switch is in the disable position, the digital control outputs should be disabled.
- A status input contacts should be available to monitor the position of this switch.

- The switch position should be reported to the SCADA/DMS system.
- The required number of I/O points should be the responsibility of the contractor, considering 20 % spare of I/O points.
- It should be possible to expand the RTU capacity by an additional 20% of the initially delivered (including spares) I/O points by providing spaces for adding cards and terminations at future date.
- STATUS INPUTS
  - The contractor should supply the necessary sensing voltage, current limiting, input isolation, and bounce filtering of all status inputs.
  - The de-bounce period for each status input should be individually configurable.
  - The input circuit of the status input modules should be optically isolated from external signal.
  - Each input circuit should include a LED indicator next to the circuit terminations to show the status of the associated input contact.
  - The state of each status point should be reported to the SCADA / ADMS on a contention basis. That is, a status point should be not being reported unless the point state has changed from the last scan.
  - The RTU should also report the state of selected status point upon receipt of a demand scan request from the SCADA / ADMS.

- CONTROL OUTPUTS
  - The RTU should include on / off device control points to support control actions initiated from SCADA / DMS.
  - The RTU should perform on / off control actions using complimentary pairs of contact outputs.
  - The RTU should include momentary control outputs as required by the feeder device being controlled.
  - The controlled outputs should be capable of driving a load of (5) amps at the primary control voltage with provision for an additional NO contact. Besides, DI status of command execute acknowledgement.
  - External auxiliary control relays are not preferred but may be applied if integral relays do not satisfy the above ratings, where these relays should be supplied by the contractor.
  - All control point should follow SCBO procedure for control operation.
  - Point selection should be cancelled automatically following the completion of the control action, and reselection of the point should be required for subsequent control actions.
- Analog Input
  - RTU should be able to capture analog values from current & voltage transformers or sensors.

5.4.7. RTU Time and Date

- The RTU should have an internal real time clock (RTC) for data collection coordination and time tagging and should include support for feeder fault detection.
- The RTU RTC should be maintained within hundred (100) millisecond of the same time reference used by the respective SCADA/DMS.
- The RTU synchronization should have at least two synchronization sources: primary through the communication protocol and secondary through SNTP protocol.

5.4.8. Diagnostic Software

- Software should provide to continuously monitor operation of the RTU and report RTU hardware errors to the SCADA/DMS.
- The software should check for memory, processor and input/output errors and failures.

5.4.9. Quality Assurance

- The remote terminal unit (RTU) package should be designed, constructed and tested in accordance with NEMA, NEC and IEC standards.
- Prior to shipment, the remote terminal unit (RTU) package should be fully functionally tested at the factory, as an assembled unit with communications devices installed and operational to insure proper system operation.
- All pilot devices and controls should be tested to verify proper operation. Testing documentation should be furnished at the engineer's request.

5.4.10. RTU Testing

- Type Test: it should comply the following standards:
  - IEC 60255- 4, 5 and 6 and IEC 60255-22.1, 2 and 3. & IEC 61000 - 4 -2 and 3.
- Routine Test: the RTU should pass the manufacture’s standard routine tests in accordance with referenced standard.
- The RTU and communication devices shall be approved by NTRA.

5.4.11. Signal List

Item	Signal Description	Signal acquisition	Type of data point
Measurements			
Per [X], [Y], & [M1]	3 phase and zero sequence MV currents	measured values	
	3 phase MV voltages	measured values	
	Active power	Calculated value	
	Reactive power	Calculated value	
	Power factor	Calculated value	
	Active Energy	Calculated value	
	Fault Currents (Ia-Ib-Ic-Io) for [Y1]	measured values	
Per [Y] Low voltage side (as option as per ….EDC requirements)	3 phase and neutral currents	measured values	
	3 phase and neutral voltages	measured values	
	Transformer Temperature	measured values	
	Active power	Calculated value	
	Reactive power	Calculated value	
	Power factor	Calculated value	
	Active Energy	Calculated value	
	Reactive Energy	Calculated value	
Alarms			
Per RMU	Gas low pressure for one tank or for each cell in case of separate tanks	Hard wired	Single point
	Gas insufficient pressure for one tank or for each cell in case of separate tanks	Hard wired	Single point
	RMU outdoor enclosure door open	Hard wired	Single point
	LV cabinet door open	Hard wired	Single point
	Energy tariff meter box door open	Hard wired	Single point
	CT for tariff meter door open	Hard wired	Single point
	Power supply failure (any fault in PS)	Comm./ hard wired	Single point
	AC power supply fail LV	Comm./ hard wired	Single point

	Auxiliary DC supply loss	Comm./ hard wired	Single point
	Alarm for low level battery	Comm./ hard wired	Single point
	Network Operator (1) Online	Communication	
	Network Operator (2) Online	Communication	
	Phase rotation (ABC)	Communication	
Per [X]	Forward Direction SC fault	Communication	
	Backward direction SC fault	Communication	
	Forward Direction EF fault	Communication	
	Backward direction EF fault	Communication	
	Cable door open	Hard wired	Single point
	Cable door open	Hard wired	Single point
Per [Y1]	Protection relay OC Trip	Communication/hardwire	
	Protection relay EF Trip	Communication/hardwire	
	Protection relay Faulty	Hard wired	Single point
	Buchholz relay alarm/trip (Oil Tr.) or Temperature alarm /trip (Dry Tr.)	Hard wired	Single point
	Cable door open	Hard wired	Single point
Per [Y2]	Any fuse blown	Hard wired	Single point
	Fuse rack out (for GIS) if available	Hard wired	Single point
	Cable door open	Hard wired	Single point
Status			
Per RMU	LBS ON/OFF	Hard wired	Double point
	CB ON/OFF	Hard wired	Double point
	ES ON/OFF	Hard wired	Double point
	Local/Remote	Hard wired	Double point
For [X] & [Y]	Lock/Unlock (Local/Remote)	Hard wired	Double point
Commands			
For [X] & [Y]	Close command	Digital Output	
	Open command	Digital Output	
	Reset alarms	Communication	

6. COMMUNICATION

6.1. Communication with SCADA

- The RTU shall have at least 2 communication ports which can be available to work in parallel. Each port shall be able to support individually the communication protocols. The communication medium and protocol shall be set from the DMS or set locally. For maintenance port, RS232, USB, or equivalent connection, as a minimum, shall be provided for interface to the RTU.
- The RTU shall be able to communicate with the SCADA on 2 channels.
- The RTU shall accept communication with at least 2 SCADA simultaneously.

6.2. Protocol

- The RTU shall comply with IEC 870-5-104 and Modbus as mandatory in addition IEC 870-5-101, IEC61850, DNP3.0 protocols if available. The RTU shall support Secure Authentication according to IEC 62351-5/3, and provide the data base as an IEC 61850 data model.
- Communication protocols of SCADA shall be supported by authentication and encryption.
- The RTU shall comply with both server mode for the SCADA and client mode for devices in the substations.

6.3. Transmission

- The communication system with the control center shall be based on 3G/4G or according to telecom study. The RTU power supply shall be sized to supply the communication modem.

6.4. Transmitted data

- The RTU shall transmit all status indications and measurement values to the SCADA system. Each data point shall be individually configurable to enable or disable transmission to SCADA.
- Status and measurement data shall be transmitted either spontaneously or periodically, based on the configured transmission method:
    - Measurements shall be transmitted according to defined thresholds or dead-band settings.
    - Status indications shall be transmitted upon state change.

7. CYBER SECURITY

- In order to secure all controls and data acquisition, the RTU shall be designed to be compliant with NERC and IEC62351 requirements. The RTU shall support secure access based on RBAC.
- RTUs and routers shall support authentication via a RADIUS or LDAP server.

- Local and remote access connection shall be secured for maintenance (locally and remotely) with HTTPS or SFTP or SSH protocols.
- The Firmware of the RTUs shall be digitally signed through embedded crypto-chip on the CPU which provides secure for uploading the firmware to the RTU.

7.1. Future proof design

- Remote firmware update
  - The RTU shall support remote firmware updates

- Centralized RBAC management
  - The RTU shall be evolutive in order to be compatible with a full centralised RBAC management in compliance with IEC 62351-8

7.2. Hardening

- Device hardening
  - Disabled or unused functionality shall not compromise security.
  - Unnecessary services and programs shall be removed. If removal is not possible, the unnecessary services and programs shall be disabled.
  - The RTU must provide mechanisms to prevent illegitimate firmware to be loaded into the system

- Interface minimization
  - Each interface shall support only the data types and protocols needed to meet the functional requirements.
  - Unused interfaces and ports shall be removed. If removal is not possible, the unused interfaces and ports shall be disabled.
  - A complete list of supported data types and supported communication protocols per interface shall be provided.
  - All hardware interfaces that are used for programming or debugging shall be completely removed after production.
- Account hardening
  - The RTU shall not contain active default, guest and anonymous accounts.
  - All remote access to root accounts on the RTU shall be disabled.
  - All Vendor-owned accounts where feasible shall be removed.
  - The list of all accounts on the RTU shall be provided.

7.3. Compliance to security standard

The RTU shall follow the IEC 62351 standards and at least:

- IEC 62351-5: 2013 or IEC 62351-3

7.4. Configuration

- Access to the RTU by configuration tool shall be possible only through secured connection: HTTPS for Webserver or SSH for console and configuration tool.

7.5. Access control (RBAC)

- The RTU shall support the implementation of Role-based Access Control in compliance with IEC 62351-8.
- The RTU should allow to manage user authentication through a Radius server or LDAP (secured via TLS) according to IEC62351-8.
- It must be possible to carry out changes by configuration files through a secure way.
- It shall be possible to assign each role individual security credentials.
- A specific tool shall permit to configure the security policy, role and password.
- It shall be possible to bind roles to individual user accounts on the RTU.
- The minimum following function and data shall be controlled through RBAC:
  - Configuration files
  - Software update
  - User management
  - Executing program or shell command
  - I/O on local maintenance access
- Management of Security passwords
  - The RTU service application shall support individual user passwords.
  - Passwords shall be stored together with a salt using an allowed cryptographic hash function.
  - The RTU service application shall enforce a high complexity of passwords.
  - The RTU shall lock the access after several password failures.
  - User Authentication
  - The RTU shall authenticate the communication parties on the WAN interface using a challenge-response protocol based on message authentication codes The RTU shall terminate the connection if the user authentication fails.
  - The RTU shall authenticate the communication parties on the Local Maintenance interface.

- It shall be possible to configure the RTU so that it blocks authentication requests, either temporarily or permanently, from an account after a number of failed login attempts. The number of failed logins attempts and the time the account is blocked shall be configurable.

- Central management of user account
  - The RTU should allow managing user authentication through a Radius or LDAP server.

7.6. Security Log

- The RTU shall provide a local audit trail for all security events that occur.
- Log files shall be produced in Syslog format.
- Security events shall be logged locally in a dedicated security log or/and on a SYSLOG server.

7.7. Security testing & Certificate

- The RTU shall have a penetration test report done by independent company.
- The bidder must follow the NTRA & EG-Cert. cyber security approval requirements and the certification for the RTU system.

7.8. Documentation

- Secured Versioning
  - All released versions (hardware, firmware, software) of a device or product shall be uniquely identifiable.
  - Exchangeable hardware modules shall be versioned separately.
  - For Asset Monitoring the RTU shall provide firmware version, Serial number, date of installation, via SNMPv3 Agent.”
- Design Documentation
  - The Protocol Implementation Conformance Statement as in IEC 62351 and IEC 60870-5-7 shall be provided on request.

8. TESTING AND INSPECTION

8.1. Type Test:

All equipment shall be successfully type tested as EEHC approved laboratory in accordance to latest relevant standard for the following:

- High-Voltage switches as per IEC62271-102/103
- High-Voltage switch fuse as per IEC62271-105
- Circuit breaker as per IEC62271-100
- Degree of protection as per IEC60529.

- High-voltages switchgear and control gear as per IEC 62271-200
- RTU, protection relays or electronic devices such as fault passage indicators should be tested according to: IEC 60255- 4, 5 and 6 and IEC 60255-21.1, 2 and 3. & IEC 61000– 4 and 6.

Note: see appendix A for the detailed type tests.

8.2. Routine Test:

Routine tests of the RMU should be carried out according to latest relevant IEC standards by the manufacturer; a representative of Distribution Company will attend these tests before acceptance.
The…...EDC have the right to add any test during delivery to check the quality of the product.

- High-Voltage switches as per IEC62271-102/103
- High-Voltage switch fuse as per IEC62271-105
- Circuit breaker as per IEC62271-100
- Measurement unit as per IEC81869-2 and 3
- Control panel and parts as per Iec60068-2-1/2
- High-voltages switchgear and control gear as per IEC 62271-200
- Each and frame equipped with protection relay a primary current injection test shall be conducted
- …EDC reserves the right to visit the factory during manufacture of any or all items covered by this specification, for inspection of material or witness of tests.

Accordingly, the manufacturer shall give …EDC testing schedule.

Panel's Type:	Panel's Name	
DESCRIPTION OF TESTS: Pass(P), Fail (F) , Un-completed (U) , Not available (NA)		
A	Electrical and Mechanical characteristics of RMU	**Evaluation**
1	Visual inspection tests: ▪ All integrated component is compliant with the drawing ▪ No physical damage ▪ Check the nameplates and general labeling form/to. ▪ Checking gas pressure indicator. ▪ Inspect MV cable compartment, earth bar, cable support, cable clamp	
2	Dielectric test MV and LV	

3	Voltage drops on Earthing circuit and main circuit	
4	▪ Manual operating tests MV breaker / switch disconnector and mechanical interlock. ▪ Electrical operating tests for circuit breaker/ switch disconnector ▪ Circuit breaker contact resistance and timing test.	
5	Interlocks The purpose of the test is to check that it is not possible to operate the functional unit when there is an interlock	
B	RTU	
1	▪ Checking the Interface with RMU through the RTU: (Monitor/control): ▪ Local / Remote management ▪ Smoke detector / Fire detector / Heat measurements	
2	▪ Compliance with the signal list	
3	▪ Synchronization and time stamping of Sequence of Event	
4	▪ Operators management / Cyber security	
C	Fault passage Indicator	
	The purpose of the test is to check the function of the FPI: ▪ Directional Phase to phase fault. ▪ Directional Phase to earth fault.	
D	Protection Relay	
1	The purpose of the test is to check the trip of the transformer breaker in case of trip condition: ▪ Phase fault ▪ Earth fault	
E	Power supply	
1	The purpose of the test is to check the operation in case of supply failure (No. of operation/ backup time)	
F	Telecommunication for Kiosks	
1	Check communication from Kiosk (with Single modem and Dual SIM) to DCC	

9. TRAINING

…. EDC reserves its right to claim a priceless training of its representatives on the vendor's factory, site, and lab.

The training should include and not limited to the following:

1. Installation.
2. Testing & Commissioning.
3. Troubleshooting.
4. Operating and Maintenance

10. AFTER SALE SERVICE

The tenderer should have a licensed service center in Egypt covers all its deliverables.

….EDC reserves its right to claim a free after sale service all over the warranty period guaranteed by the tenderer that covers all its deliverables, after sale services should include and not limited to the following:

1. Technical support.
2. Maintenance.
3. Original Spare parts.
4. Submittals.

- All the services, spare parts, materials and accessories will be needed during the after-sale maintenance should be guaranteed by the tenderer immediately and wherever required (Warehouse – Site – Lab. …. etc.)

11. WARRANTEE

- The supplier guarantees the SRMU and RTU against all defects arising out of faulty design or workmanship, or of defective material for period of (5) years from date of delivery.
- Warranty period for gas tightness (seal pressure system) shall conform to the relevant IEC and the contractor shall assume full responsibility for no gas leakage during the service life.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.

12. MARKING

- The following data should be engraved or laser printed on an aluminum/stainless steel nameplate riveted on the Ring main unit and Enclosure and located in a visible and legible place when the switchboard is installed showing the following information:
  - Manufacturer's name
  - Type/Model
  - Serial Number and year of Manufacturing
  - Distribution company name
  - order number
  - indoor or outdoor application
  - Rated voltage and rated insulation level
  - Rated frequency
  - N1 no. LBS + N2 no. CB/F and the type (GIS/AIS)
  - Rated and duration of short-circuits current (KA/sec)
  - Making current KA
  - degree of protection
  - Gross Weight(KG)
  - IEC standards
  - IAC type and (KA/time).
- The following data should be cleared per cell (Switch disconnector, Switch fuse disconnector/CB) and located in a visible and legible place showing the following information, Entries on the plate should be indelibly marked be etching or engraving:
  - Type/Model
  - Rated voltage and rated insulation level
  - IEC standards
  - Rated current (A) and Rated voltage (V).
  - Rated and duration of short-circuits current (KA/sec)
  - Making current KA
  - Electrical and mechanical endurance
  - Rated frequency
  - Serial Number and year of Manufacturing

13.SUBMITTALS

Vendor shall provide the following with the quotation:

Clause by clause compliance with the specification.

- All necessary drawings including the scheme drawing of the RMU, dimensions, single line diagram of (control circuit, tripping circuit, closing circuit, voltage supply circuit, VT and CT connections, Alarms, communication……etc.)
- Technical catalogues containing the technical data for all offered components of the unit should be submitted.
- Installation and maintenance instructions.
- Copy of type test report.
- A valid approved letter must be submitted from the Egyptian Electricity Holding Company
- A comprehensive list of manufacturer's recommended spare parts. The quantities offered should be adequate for the initial 5 years of operation. Firm price and delivery period shall be quoted for each item and …. EDC has the right to request the spare part list or not
- Vendor shall provide the following after signing of purchase order:
    - Details of manufacturing and testing schedules
    - Routine test reports (including tightness test)
- Vendor shall provide the following with delivering:
    - Operation (user) manual.
    - Handles, Master Keys, Locks…. for operation and locking.
    - External device to check the phases concordances per order.
    - Testing rods for HIPOT (High Potential) test per order.
    - Testing rods for current injection test per order.

14.OPTION LIST:

The following table should be filled/attached by …EDC during tendering.

NO.	Option	Needed	Not Needed
1	Ambient temperature: 50°C		
2	Outdoor RMU		
3	The RMU may be extendable		
4	The internal arc class of the RMU should be provided with downward/up ventilation		
5	A suitable base for RMU shall		
6	The enclosure should be hot-dip galvanized (HDG)		
7	Sensitive earth fault for protection relay		
8	Every load break switch to be on a separate tank		
9	The battery hours sizing		
10	The RMU unit may be provided with a separate Local/Remote switch on the LV Box		
11	Lock/Unlock (for each cell)		
12	Two communication modems		
13	Two separate batteries for RMU operation and RMU automation devices		
14	Interlock (cannot close the earth switch if the cable energized)		
15	Low Voltage Measurements		
16	Circuit breaker rated current (200-400-630)A	………………………….….	
17	[X], [Y1 or Y2], and/or [M1 or M2] should be defined according to ….EDC requirements: NO. of [X…]: , NO. of [Y1…]: NO. of [Y2…]: ,NO. of [M1…]: NO. of [M2…]: H.R.F capacity: ,C.T. Ratio:		

15.TECHNICAL DATA SCHEDULE

The guarantee schedules should be submitted and the tenderer should fill in the attached guarantee tables. ….EDC reserves the right to reject any tender not accompanied with clear and complete guarantee tables

GUARANTEE TABLE

No	Item	Vendor Spcs.
General		
1	Rated (Maximum operating) voltage	…….KV
2	Nominal operating (Service) voltage	…….KV
3	Power frequency R.M.S. withstand voltage (1 min)	…….KV
4	Impulse withstand voltage (1.2/50) µs peak	…….KV
5	Rated frequency	…….HZ
6	Color	…….
7	Auxiliary voltage	…….V
8	IP Class (Gas Tank)	…….
9	Copper / Aluminum purity	……%
10	Copper / Aluminum Conductivity	……MS/m
Environmental conditions		
1	Ambient temperature	……°C
2	Average relative humidity	……°C
3	Maximum altitude	……m

Design and construction requirements		
Indoor RMU		
1	RMU Type	……
2	RMU configuration	……
3	Thickness of sheet steel/IK protection	……mm
4	Thickness of electrostatic paints	…… microns
5	Dimensions: Width *Depth *Height	…..mm*…...mm*…...mm
6	The internal arc: Class: KA: Time: Outlet:	 …… ……KA ……sec ……
7	Load Break Switch:	
8	IEC Standards	
Switch disconnector		
1	Rated Voltage	……KV
2	Rated normal current at 40&45&50 C°	……/……/…… A
3	Short-time withstand current	……KA
4	Making capacity	……KA
5	Mechanical endurance	……
6	Electrical endurance	……
Earth Switch for Switch disconnector		
1	Short-time withstand current	……KA
2	Making capacity	……KA
3	Mechanical endurance	……

4	Electrical endurance	……
Switch fuse disconnector		
1	Rated Voltage	……KV
2	Rated normal current at 40&45&50 C°	……/……/…… A
3	Short-time withstand current	……KA
4	Making capacity	……KA
5	Mechanical endurance	……
6	Electrical endurance	……
Earth Switch for Switch Fuse disconnector		
1	Short-time withstand current	……KA
2	Making capacity	……KA
3	Mechanical endurance	……
4	Electrical endurance	……
Circuit breaker		
1	Type/Model	……/……
2	Interruption medium	……
3	Rated Voltage	……KV
4	Rated normal current at 40&45&50 C°	……/……/…… A
5	Short-time withstand current	……KA
6	Making capacity	……KA
7	Mechanical endurance	……
8	Electrical endurance	……
9	Duty cycle	……
10	Opening time	……ms
11	Operating mechanism	……
Earth Switch for Circuit breaker		

1	Short-time withstand current	……KA
2	Making capacity	……KA
3	Mechanical endurance	……
4	Electrical endurance	……
	CT	
1	Product type	……
2	Highest voltage for equipment Um (R.M.S)	……KV
3	Rated primary service voltage	……KV
4	Rated frequency	……HZ
5	Rated primary current	……A
6	Rated secondary current	……A
7	Current continuous factor	……
8	Rated short time "thermal" withstand current (RMS) For 1 sec (KA)	……KA
9	Accuracy class	……
10	Rated Burden	……VA
11	Creepage distance	……cm/kV
12	Class of insulation	……
	VT	
1	Product type	……
2	Highest voltage for equipment Um (rms)	……KV
3	Rated primary service voltage	……KV
4	Rated frequency	……HZ
5	Rated primary voltage	……V
6	Rated secondary voltage	……V

7	Voltage factor	……
8	Rated impulse withstand voltage for primary windings	……KV
9	Short duration withstands (rms) voltage for primary windings (1min)	……KV
10	Rated power frequency short duration withstands for secondary winding for 1 minute	……KV
11	Accuracy class	……
12	Rated Burden	……VA
13	Creepage distance	……cm/kV
14	Class of insulation	……
Current Sensors or Ring core CT		
1	Rated current	……A
2	Rated Burden	……VA
3	Rated short time "thermal" withstand current (RMS) For 3 sec I_{th}	……KA
4	Accuracy class	……
5	Accuracy limit factor	……
Voltage Detection and Indicating System VDIS		
1	Product type	……
2	IEC standards	……
3	It shall have a low voltage hot phasing facility. The lamps / LCD shall be powered by bushing-type capacitive dividers.	Yes/no
4	It shall detect the voltage presence /absence and be fitted with a 3 LED/LCD.	Yes/no
5	It shall deliver the measured signal to the RTU.	Yes/no
Interlocks		

1	Matching all requirements	Yes/no
	Enclosure Outdoor Use:	
1	Thickness of sheet steel/IK protection	……mm
2	Thickness of electrostatic paints	……microns
3	Dimensions: Width *Depth *Height	……mm*……mm*……mm
4	IP protection	……
	Fault passage Indicator (FPI)	
1	Integrated/separate	…….
2	Feeding source	…….
3	IP	…….
4	Matching all requirements	Yes/no
	Capacitor divider and phase comparator	
1	Every load break switch or circuit breaker cell should be equipped with three capacitor dividers connected to the voltage presence indicator system monitored by LEDs or LCD to indicate the voltage presence on the cable.	Yes/no
2	There should be a facility to check the phases concordances with the use of external device. This external device shall be provided with the RMU.	Yes/no
	GIS	
1	Model	…….
2	IP protection for Cable Box and Mechanism	…….
3	C.S.A of earthing bar	…….
4	Interface type of Terminal bushings and size of bolt	…….
5	Stages No. for the gas pressure indication for the tank	…….
6	Rated short circuit current for 1 Sec	…….

AIS		
1	Model	…….
2	IP protection	…….
3	C.S.A of Bus bar	…….
4	C.S.A of earthing bar	…….
5	Creepage distance	…….cm/kV
6	Heater of 100-watt, 220v AC with a hygrostat and a M.C.B per each module.	Yes/no
7	Additional 100 watt, 220v AC heater shall be installed in the bus bar compartment with its associated MCB and hygrostat.	Yes/no
8	Rated short circuit current for 1 Sec	
Protection Relay for [Y1] module		
1	Maker 's name	…….
2	Country of manufacture	…….
3	Type and designation	…….
4	Standard specification with which relay complies	…….
5	Relays characteristics	…….
6	Range of relay setting	I……(A)/ T……(Sec)
7	Current rating of relay coil	…….A
8	Voltage rating of relay coil	……. V
9	Auxiliary supply rang	…….DC (V)
10	Relay order number	…….
11	Dual power supply	Yes / No

12	Low power trip coil	Yes / No
13	Standard reconfiguration functionality	…….
14	External trip function	Yes / No
15	Communication protocol / port	……./…….
16	The relay should meet the latest revision of IEC publication 60255 standard	Yes / No
	RTU automation	
A	Low Voltage Enclosure Box	
1	IP protection	…….
2	Thickness of sheet steel	…….mm
3	Operating temperature range	…….°C to…….°C
4	It should be Pad lockable and should be stand on the top of the RMU	Yes/no
B	Power Supply (PS) and battery	
1	Input voltage /output voltage	…….v
2	Rated current	…….A
3	Power supply capacity	…….
4	Battery type	…….
5	Battery sizing (houres)	…….H
6	No of operation cycles for the battery	…….
7	Operating temperature of the PS and battery	…….°C
8	The output DC voltage shall be constant with ripple voltage less than 1 % without battery.	
C	Remote Terminal Unit (RTU):	
1	Compliance with communication protocol and ports	Yes or No
2	Sequence of event resolution 1 msec/ No. of events	Yes or No/…….

3	Operating temperature of the RTU	…C°.
4	Compliance with No. of spare input and output points	Yes or No
D	Signal list	
1	Compliance with signal list sheet	Yes/no
E	Communication	
1	The 3G/4G Router shall support LTE with bands (B1, B2, B8, B20, and B28).	Yes/no
2	No. of SIM cards per router	……..
F	Cyber Security	
1	The RTU shall follow the IEC 62351 standards and at least: ▪ IEC 62351-5 : 2013 ▪ IEC 62351-3	Yes/no
2	The RTU support secure access based on role-based access control.	Yes/no
3	The RTU support remote firmware updates	
4	The RTU support HTTPS or SFTP or SSH protocols for Local and remote access connection.	

We guarantee the data given above for the equipment offered.

Signature: ………………………………

Date: ……………………………………

APPENDIX A

RMU (AIS / GIS) type tests

1. High-Voltage switches as per IEC62271-103/102 (LBS)

1. Dielectric tests.
2. Temperature-rise tests.
3. Measurement of the resistance circuit.
4. Short-time withstand current and peak withstand current tests.
5. Tightness tests.
6. Making and breaking test.
7. Mechanical test.

2. High-Voltage switch fuse as per IEC62271-105 (fuse-LBS):

1. Dielectric tests
2. Temperature-rise tests.
3. Measurement of the resistance of circuit.
4. Tightness tests.
5. Tests to prove the combination to make and break the specified currents.
6. Mechanical operation test
7. Short-time withstand current and peak withstand current tests.

In case of the switch fuse disconnector has the same design of switch disconnector, Type tests no. 5, 7 and 8 shall be only applied.

3. Circuit breaker as per IEC62271-100:

1. Dielectric tests
2. Measurement of the resistance of the main circuit
3. Temperature-rise tests
4. Short-time withstand current and peak withstand current tests
5. Mechanical operation test at ambient temperature (class M1)
6. Short-circuit current making and breaking tests
7. Tightness test
8. Electrical endurance tests (only for Ur ≤ 52kV). See Note 5.
9. Capacitive current switching tests.

4. <u>High-Voltages switchgear and control gear as per IEC62271-200:</u>

1. Rated Insulation level.
2. Measurement of the resistance of circuits
3. Rated continues current and measurement of the resistance of circuits
4. Short-time withstand current and peak withstand current tests
5. Making and breaking capacity of the included switching device. <u>See Note 1</u>
6. Mechanical operation of the included switching devices and removable parts. <u>See Note 2</u>
7. IP protection code.
8. IK Protection degree
9. Pressure withstand of gas - filled compartments.
10. Tightness of gas – or liquid- filled compartments. <u>See Note 3.</u>
11. Internal Arc Test
12. Electromagnetic compatibility (EMC). <u>See Note 4</u>.

<u>Note 1 according to IEC 62271-200:</u>

Switching devices forming part of the main circuit and earthing switches of assemblies shall be tested to verify their rated making and breaking capacities according to the relevant standards and under the proper conditions of installation and use. That is, they shall be tested as normally installed in the assembly with all associated components that may influence the performance, such as connections, supports, provisions for venting. These tests are not necessary if making and breaking tests have been performed on the switching devices installed in assembly with identical or more onerous conditions.

<u>Note 2 according to IEC 62271-200:</u>

All switching devices not previously tested as mounted in the assembly, shall be operated 50 times C-O, mounted in the assembly. Test conditions and criteria to pass the test are identical to the ones defined on each corresponding switching device standard for mechanical tests

<u>Note 3 according to IEC 62271-1:</u>

Where possible, the tests should be performed on a complete system. If this is not practical, the test may be performed on parts, components or subassemblies.

Note 4 according to IEC 62271-1:

Auxiliary and control circuits of switchgear and control gear shall be subjected to electromagnetic emission/immunity tests if they included electronic equipment or components in other cases no tests are required

Note 5 according to IEC 62271-100:

The electrical endurance capability of circuit-breakers, that are not for auto-reclosing duty, is demonstrated by performing the terminal fault test-duties of 7.107 without intermediate maintenance. Additional tests are not required.